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LDMOS transistor and method of forming the same

Abstract

A MOS transistor includes a substrate, a first region, a second region, a source region, a drain region, an active gate stack, and a dummy gate stack. The substrate has a first conductivity. The first region having the first conductivity is formed in the substrate. The second region having a second conductivity is formed in the substrate and is adjacent to the first region. The source region with the second conductivity is formed in the first region. The drain region with the second conductivity is formed in the second region. The active gate stack is disposed on the first region. The dummy gate stack is disposed on the second region, and the dummy gate stack is electrically coupled to a variable voltage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional application of and claims the priority benefit of U.S. application Ser. No. 17/028,998, filed on Sep. 22, 2020. The prior application Ser. No. 17/028,998 is a continuation application of and claims the priority benefit of U.S. application Ser. No. 15/054,078, filed on Feb. 25, 2016. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

(1) MOSFETs are voltage controlled devices. When a control voltage greater than the threshold of the MOSFET is applied to the gate of a MOSFET, a conductive channel is built between a drain and a source of the MOSFET. On the other hand, when the control voltage is less than the threshold of the MOSFET, the MOSFET is turned off accordingly. Currently, MOSFETs are divided into three sub-categories, planar MOSFETs, lateral double diffused MOS (LDMOS) FETs and vertical double diffused MOSFETs. In comparison with other MOSFETs, the LDMOS is capable of delivering more current per unit area because its asymmetric structure provides a short channel between the drain and the source of the LDMOS. In order to increase the breakdown voltage of the

LDMOS, the gate poly-silicon of the LDMOS may be extended to make an overlap with the drift region of the LDMOS. Such an overlap functions as a field plate to maintain the breakdown voltage of the LDMOS.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a top view of a LDMOS transistor in accordance with some embodiments.
- (3) FIG. 2 is a cross-sectional view of a LDMOS transistor in accordance with some embodiments.
- (4) FIG. 3 is a top view of a LDMOS transistor with contact structures in accordance with some embodiments.
- (5) FIG. 4 is a cross-sectional view of a LDMOS transistor with contact structures in accordance with some embodiments.
- (6) FIG. 5 is a cross-sectional view of a LDMOS transistor in accordance with some embodiments.
- (7) FIG. 6 is a top view of a LDMOS transistor in accordance with some embodiments.
- (8) FIG. 7 is a cross-sectional view of a LDMOS transistor in accordance with some embodiments.
- (9) FIG. 8 is a top view of a LDMOS transistor with contact structures in accordance with some embodiments.
- (10) FIG. 9 is a cross-sectional view of a LDMOS transistor with contact structures in accordance with some embodiments.
- (11) FIG. 10 is a cross-sectional view of a LDMOS transistor in accordance with some embodiments.

DETAILED DESCRIPTION

- (12) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (14) The present disclosure will be described with respect to some embodiments in a specific context, a high voltage lateral double diffused metal oxide semiconductor (LDMOS) transistor. The embodiments of the disclosure may also be applied to a variety of metal oxide semiconductor field effect transistors (MOSFETs). In accordance with one embodiment of the disclosure, the first

conductivity is p-type and the second conductivity is n-type. In accordance with another embodiment of the disclosure, the first conductivity is n-type and the second conductivity is p-type. (15) FIG. 1 is a top view of a LDMOS transistor in accordance with some embodiments and FIG. 2 is a cross-sectional view of the LDMOS transistor of FIG. 1. Referring to FIG. 1 and FIG. 2, the LDMOS includes a substrate **102**, a first region **106**, a second region **108**, a source region **112**, a drain region **114**, an active gate stack **120**, and a dummy gate stack **122**. In some embodiments, the LDMOS further includes an isolator **104**.

(16) The substrate **102** comprises a crystalline silicon substrate (e.g., wafer). In some alternative embodiments, the substrate **102** may be made of some other suitable elemental semiconductor, such as diamond or germanium; a suitable compound semiconductor, such as gallium arsenide, silicon carbide, indium arsenide, or indium phosphide; or a suitable alloy semiconductor, such as silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. The substrate **102** may be a p-type substrate configured for an n-type LDMOS or an n-type substrate configured for a p-type LDMOS. In some embodiments, the substrate **102** may be doped with p-type dopants, such as boron or BF₃; or n-type dopants, such as phosphorus or arsenic. In order to illustrate the present disclosure, an n-type LDMOS is drawn in the figures as an example, the substrate **102** for the n-type LDMOS is a p-type substrate, and the present disclosure does not limit thereto.

(17) The first region **106** is formed in the substrate **102**. In accordance with some embodiments, the first region **106** is formed by implanting p-type doping materials such as boron, gallium, aluminum, indium, combinations thereof, or the like. In one embodiment, a p-type material such as boron may be implanted to a doping density of about $10^{15}/\text{cm}^3$ to $10^{18}/\text{cm}^3$. The first region **106** can be a P⁻ region and may be referred as a channel region.

(18) The second region **108** is formed in the substrate **102** and is adjacent to the first region **106**. In accordance with some embodiments, the second region **108** is formed by implanting n-type doping materials such as phosphorous, arsenic, nitrogen, antimony, combination thereof, or the like. In one embodiment, an n-type material such as phosphorous may be implanted to a doping density of about $10^{15}/\text{cm}^3$ to $10^{18}/\text{cm}^3$. The second region **108** can be an N⁻ region and may be referred to as an extended drift region.

(19) The source region **112** is formed in the first region **106**. In one embodiment, the source region **112** may be formed by implanting an n-type dopant, such as phosphorous at a concentration of between about $10^{19}/\text{cm}^3$ and about $10^{20}/\text{cm}^3$. The source region **112** can be an N⁺ region.

(20) The drain region **114** is formed in the second region **108**. In one embodiment, the drain region **114** may be formed by implanting an n-type dopant, such as phosphorous at a concentration of between about $10^{19}/\text{cm}^3$ and about $10^{20}/\text{cm}^3$. The drain region **114** can also be an N⁺ region.

(21) The isolator **104** is used to isolate active regions so as to prevent leakage current from flowing between adjacent active regions. The isolator **104** may include silicon oxide, silicon nitride, silicon oxynitride, or a low-K dielectric material. The isolator **104** may be formed by a thermally growing process or by a depositing process such as high-density-plasma chemical vapor deposition (HDP-CVD). In this embodiment, the isolator **104** may be fabricated by a shallow trench isolation (STI) process.

(22) The active gate stack **120** is disposed on the first region **106**. The active gate stack **120** comprises a gate dielectric layer **120a** and a gate electrode layer **120b**. In some embodiments, the gate dielectric layer **120a** may include silicon oxide, silicon nitride, silicon oxy-nitride, or high-k dielectrics. High-k dielectrics comprise metal oxides. Examples of metal oxides used for high-k dielectrics include oxides of Li, Be, Mg, Ca, Sr, Sc, Y, Zr, Hf, Al, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, and/or mixtures thereof. The gate dielectric layer **120a** may be formed using a suitable process such as atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), thermal oxidation, UV-ozone oxidation, or combinations thereof.

The gate electrode layer **120b** is disposed on the gate dielectric layer **120a**. In some embodiments, the gate electrode layer **120b** may comprise a single layer or multi-layered structure. In some embodiments, the gate electrode layer **120b** may comprise poly-silicon or metal, such as Al, Cu, W, Ti, Ta, TiN, TiAl, TiAlN, TaN, NiSi, CoSi, other conductive materials with a work function compatible with the substrate material, or combinations thereof. The gate electrode layer **120b** may be formed using a suitable process such as ALD, CVD, PVD, plating, or combinations thereof. In accordance with some embodiments, the active gate stack **120** may further comprise spacers on sidewalls of the gate dielectric layer **120a** and the gate electrode layer **120b**.

(23) The dummy gate stack **122** is disposed on the second region **108**. The dummy gate stack **122** comprises a gate dielectric layer **122a** and a gate electrode layer **122b**. In some embodiments, the gate dielectric layer **122a** may include silicon oxide, silicon nitride, silicon oxy-nitride, or high-k dielectrics. High-k dielectrics comprise metal oxides. Examples of metal oxides used for high-k dielectrics include oxides of Li, Be, Mg, Ca, Sr, Sc, Y, Zr, Hf, Al, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, and/or mixtures thereof. The gate dielectric layer **122a** may be formed using a suitable process such as atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), thermal oxidation, UV-ozone oxidation, or combinations thereof. The gate electrode layer **122b** is disposed on the gate dielectric layer **122a**. In some embodiments, the gate electrode layer **122b** may comprise a single layer or multi-layered structure. In some embodiments, the gate electrode layer **122b** may comprise poly-silicon or metal, such as Al, Cu, W, Ti, Ta, TiN, TiAl, TiAlN, TaN, NiSi, CoSi, other conductive materials with a work function compatible with the substrate material, or combinations thereof. The gate electrode layer **122b** may be formed using a suitable process such as ALD, CVD, PVD, plating, or combinations thereof. In accordance with some embodiments, the active gate stack **122** may further comprise spacers on sidewalls of the gate dielectric layer **122a** and the gate electrode layer **122b**.

(24) In one embodiment, the active gate stack **120** and the dummy gate stack **122** are formed with the same photomask process. Therefore, a material of the gate dielectric layer **120a** is the same with a material of the gate dielectric layer **122a**, and a material of the gate electrode layer **120a** is the same with a material of the gate electrode layer **122b**. A thickness of the gate dielectric layer **120a** is substantially equal to a thickness of the gate dielectric layer **122a**, and a thickness of the gate electrode layer **120a** is substantially equal to a thickness of the gate electrode layer **122b**.

(25) In accordance with some embodiments, a width **W2** of the dummy gate stack **122** is smaller than a width **W1** of the active gate stack **120**. The width **W2** of the dummy gate stack **122** ranges from 40 nm to 130 nm, for example, and the width **W1** of the active gate stack **120** ranges from 500 nm to 10 μ m, for example. In one embodiment, the active gate stack **106** overlaps with the first region **106** and does not overlap with the second region **108**, but the disclosure does not limit thereto.

(26) The active gate stack **120** is electrically coupled to a bias voltage for a LDMOS transistor, and the dummy gate stack **122** is electrically coupled to a variable voltage. The variable voltage applied to the dummy gate stack **122** may adjust a resistance of the second region **108** underneath the dummy gate stack **122**. Thus, a breakdown voltage (**BV_{dss}**) of the LDMOS transistor can be adjusted for different demands. The breakdown voltage (**BV_{dss}**) is a voltage at which currents start to flow between the source region and the drain region. Since the dummy gate stack **122** is electrically coupled to a variable voltage, the resistance of the LDMOS transistor is variable, such that it is flexible for designers to apply different breakdown voltage (**BV_{dss}**) for the single LDMOS transistor.

(27) FIG. 3 is a top view of a LDMOS transistor with contact structures in accordance with some embodiments. FIG. 4 is a cross-sectional view of a LDMOS transistor of FIG. 3. After forming the active gate stack **120** and the dummy gate stack **122**, as shown in FIG. 3 and FIG. 4, an insulating layer **130** is formed over the source **112**, the drain **114**, the isolator **104**, the active gate stack **120** and the dummy gate stack **122**. The insulating layer **130** may include silicon oxide, silicon nitride,

silicon oxynitride, a spin-on dielectric material, or a low-K dielectric material. The insulating layer **130** may be formed by high-density-plasma chemical vapor deposition (HDP-CVD), sub-atmospheric CVD (SACVD) or by spin-on. Contact structures **132**, **134**, **136**, **138** are formed in the insulating layer **130** and may be electrically connected to the source region **112**, the drain region **114**, the active gate stack **120** and the dummy gate stack **122** respectively. The contact structure **132** may be referred as a source contact, the drain structure **134** may be referred as a drain contact, the contact structure **136** may be referred as a gate contact, and the contact structure **138** may be referred as a dummy gate contact.

(28) FIG. 5 is a cross-sectional view of a LDMOS transistor in accordance with some embodiments. The LDMOS transistor shown in FIG. 5 is a modified embodiment of the LDMOS transistor in FIG. 2. For details, in the LDMOS transistor shown in FIG. 2, the active gate stack **120** overlaps with the first region **106** and does not overlap with the second region **108**. However, in the LDMOS transistor shown in FIG. 5, the active gate stack **120** partially overlaps with the first region **106** and partially overlaps with the second region **108**.

(29) FIG. 6 is a top view of a LDMOS transistor in accordance with some embodiments. FIG. 7 is a cross-sectional view of a LDMOS transistor of FIG. 6. Referring to FIG. 6 and FIG. 7, the LDMOS includes a substrate **102**, a first region **106**, a second region **108**, a third region **110**, a first source region **112a**, a drain region **114**, a second source region **112b**, a first active gate stack **120**, a second active gate stack **126**, a first dummy gate stack **122**, and a second dummy gate stack **124**. In some embodiments, the LDMOS further includes an isolator **104**.

(30) The substrate **102** comprises a crystalline silicon substrate (e.g., wafer). In some alternative embodiments, the substrate **102** may be made of some other suitable elemental semiconductor, such as diamond or germanium; a suitable compound semiconductor, such as gallium arsenide, silicon carbide, indium arsenide, or indium phosphide; or a suitable alloy semiconductor, such as silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. The substrate **102** may be a p-type substrate configured for an n-type LDMOS or an n-type substrate configured for a p-type LDMOS. In some embodiments, the substrate **102** may be doped with p-type dopants, such as boron or BF₂; or n-type dopants, such as phosphorus or arsenic. In order to illustrate the present disclosure, an n-type LDMOS is drawn in the figures as an example, and the substrate **102** for the n-type LDMOS is a p-type substrate.

(31) The first region **106** is formed in the substrate **102**. In accordance with some embodiments, the first region **106** is formed by implanting p-type doping materials such as boron, gallium, aluminum, indium, combinations thereof, or the like. In some embodiments, a p-type material such as boron may be implanted to a doping density of about $10^{15}/\text{cm}^3$ to $10^{18}/\text{cm}^3$. The first region **106** can be a P⁻ region and may be referred as a channel region.

(32) The second region **108** is formed in the substrate **102** and is adjacent to the first region **106**. In accordance with some embodiments, the second region **108** is formed by implanting n-type doping materials such as phosphorous, arsenic, nitrogen, antimony, combination thereof, or the like. In one embodiment, an n-type material such as phosphorous may be implanted to a doping density of about $10^{15}/\text{cm}^3$ to $10^{18}/\text{cm}^3$. The second region **108** can be an N⁻ region and may be referred to as an extended drift region.

(33) The third region **110** is formed in the substrate **102** and is adjacent to the second region **108**, such that the second region **108** is formed between the first region **106** and the third region **110**. In accordance with some embodiments, the third region **110** is formed by implanting p-type doping materials such as boron, gallium, aluminum, indium, combinations thereof, or the like. In one embodiment, a p-type material such as boron may be implanted to a doping density of about $10^{15}/\text{cm}^3$ to $10^{18}/\text{cm}^3$. The third region **110** can be a P⁻ region and may also be referred as a channel region.

(34) The first source region **112a** is formed in the first region **106**. In one embodiment, the first source region **112a** may be formed by implanting an n-type dopant, such as phosphorous at a

concentration of between about $10^{19}/\text{cm}^3$ and about $10^{20}/\text{cm}^3$. The first source region **112a** can be an N⁺ region.

(35) The drain region **114** is formed in the second region **108**. In one embodiment, the drain region **114** may be formed by implanting an n-type dopant, such as phosphorous at a concentration of between about $10^{19}/\text{cm}^3$ and about $10^{20}/\text{cm}^3$. The drain region **114** can also be an N⁺ region.

(36) The second source region **112b** is formed in the third region **110**. In one embodiment, the second source region **112b** may be formed by implanting an n-type dopant, such as phosphorous at a concentration of between about $10^{19}/\text{cm}^3$ and about $10^{20}/\text{cm}^3$. The second source region **112b** can also be an N⁺ region.

(37) The isolator **104** is used to isolate active regions so as to prevent leakage current from flowing between adjacent active regions. The isolator **104** may include silicon oxide, silicon nitride, silicon oxynitride, or a low-K dielectric material. The isolator **104** may be formed by a thermally growing process or by a depositing process, such as high-density-plasma chemical vapor deposition (HDP-CVD). In this embodiment, the isolator **104** may be fabricated by a shallow trench isolation (STI) process.

(38) The first active gate stack **120** is disposed on the first region **106**, and the second active gate stack **126** is disposed on the third region **110**. The first active gate stack **120** comprises a gate dielectric layer **120a** and a gate electrode layer **120b**, and the second active gate stack **126** comprises a gate dielectric layer **126a** and a gate electrode layer **126b**. The first dummy gate stack **122** is disposed on the second region **108**, and the second dummy gate stack **124** is also disposed on the second region **108**. The first dummy gate stack **122** comprises a gate dielectric layer **122a** and a gate electrode layer **122b**, and the second dummy gate stack **124** comprises a gate dielectric layer **124a** and a gate electrode layer **124b**.

(39) In some embodiments, the gate dielectric layers **120a**, **122a**, **124a**, **126a** may include silicon oxide, silicon nitride, silicon oxy-nitride, or high-k dielectrics. High-k dielectrics comprise metal oxides. Examples of metal oxides used for high-k dielectrics include oxides of Li, Be, Mg, Ca, Sr, Sc, Y, Zr, Hf, Al, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, and/or mixtures thereof. The gate dielectric layers **120a**, **122a**, **124a**, **126a** may be formed using a suitable process such as atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), thermal oxidation, UV-ozone oxidation, or combinations thereof. In some embodiments, the gate electrode layers **120b**, **122b**, **124b**, **126b** may respectively comprise a single layer or multi-layered structure. In some embodiments, the gate electrode layer **120b** may comprise poly-silicon or metal, such as Al, Cu, W, Ti, Ta, TiN, TiAl, TiAlN, TaN, NiSi, CoSi, other conductive materials with a work function compatible with the substrate material, or combinations thereof. The gate electrode layers **120b**, **122b**, **124b**, **126b** may be formed using a suitable process such as ALD, CVD, PVD, plating, or combinations thereof. In accordance with some embodiments, the first active gate stack **120** may further comprise spacers on sidewalls of the gate dielectric layer **120a** and the gate electrode layer **120b**, the second active gate stack **126** may further comprise spacers on sidewalls of the gate dielectric layer **126a** and the gate electrode layer **126b**, the first dummy gate stack **122** may further comprise spacers on sidewalls of the gate dielectric layer **122a** and the gate electrode layer **122b**, and the second dummy gate stack **124** may further comprise spacers on sidewalls of the gate dielectric layer **124a** and the gate electrode layer **124b**.

(40) In one embodiment, the first active gate stack **120**, the second active gate stack **126**, the first dummy gate stack **122** and the second dummy gate stack **124** are formed with the same photomask process. Therefore, the materials of the gate dielectric layer **120a**, **122a**, **124a**, **126a** are the same, and the materials of the gate electrode layer **120a**, **122b**, **124b**, **126b** are the same. The thicknesses of the gate dielectric layers **120a**, **122a**, **124a**, **126a** are substantially identical, and the thicknesses of the gate electrode layers **120b**, **122b**, **124b**, **126b** are substantially identical.

(41) In accordance with some embodiments, a width W2 of the first dummy gate stack **122** is

smaller than a width W1 of the first active gate stack **120**. The width W2 of the first dummy gate stack **122** ranges from 40 nm to 130 nm, for example, and the width W1 of the first active gate stack **120** ranges from 500 nm to 10 μ m, for example. A width W3 of the second dummy gate stack **124** is smaller than a width W4 of the second active gate stack **126**. The width W3 of the second dummy gate stack **124** ranges from 40 nm to 130 nm, for example, and the width W4 of the second active gate stack **126** ranges from 500 nm to 10 μ m, for example. In one embodiment, the first active gate stack **120** overlaps with the first region **106** and does not overlap with the second region **108**, but the disclosure does not limit thereto. The second active gate stack **126** overlaps with the third region **110** and does not overlap with the second region **108**, but the disclosure does not limit thereto.

(42) The first active gate stack **120** is electrically coupled to a first bias voltage for a LDMOS transistor, and the second active gate stack **126** is electrically coupled to a second bias voltage for the LDMOS transistor. The first bias voltage can be the same to the second bias voltage or different from the second bias voltage. The first dummy gate stack **122** is electrically coupled to a first variable voltage, and the second dummy gate stack **124** is electrically coupled to a second variable voltage. The first variable voltage can be the same to the second variable voltage or different from the second variable voltage. The first variable voltage applied to the first dummy gate stack **122** and the second variable voltage applied to the second dummy gate stack **124** may adjust a resistance of the second region **108** underneath the first dummy gate stack **122** and a resistance of the second region **108** underneath the second dummy gate stack **124**. Thus, a breakdown voltage (BV_{dss}) of the LDMOS transistor can be adjusted for different demands. Since the first dummy gate stack **122** is electrically coupled to the first variable voltage and the second dummy gate stack **124** is electrically coupled to the second variable voltage, the resistance of the LDMOS transistor is variable. It is flexible for designers to apply different breakdown voltage (BV_{dss}) for the LDMOS transistor.

(43) FIG. **8** is a top view of a LDMOS transistor with contact structures in accordance with some embodiments. FIG. **9** is a cross-sectional view of a LDMOS transistor of FIG. **8**. After forming the first active gate stack **120**, the second active gate stack **126**, and the first dummy gate stack **122** and the second dummy gate stack **124**, as shown in FIG. **6** and FIG. **7**, an insulating layer **130** is formed over the first source **112a**, the second source **112b**, the drain **114**, the isolator **104**, the first active gate stack **120**, the second active gate stack **126**, the first dummy gate stack **122** and the second dummy gate stack **124**. The insulating layer **130** may include silicon oxide, silicon nitride, silicon oxynitride, a spin-on dielectric material, or a low-K dielectric material. The insulating layer **130** may be formed by high-density-plasma chemical vapor deposition (HDP-CVD), sub-atmospheric CVD (SACVD) or by spin-on. Contact structures **132a**, **132b**, **134**, **136a**, **136b**, **138a**, **138b** are formed in the insulating layer **130** and may be electrically connect to the first source region **112a**, the second source region **112b**, the drain region **114**, the first active gate stack **120**, the second active gate stack **126**, the first dummy gate stack **122** and the second dummy gate stack **124**, respectively. The contact structures **132a**, **132b** may be referred as source contacts, the drain structure **134** may be referred as a drain contact, the contact structures **136a**, **136b** may be referred as gate contacts, and the contact structures **138a**, **138b** may be referred as dummy gate contacts.

(44) FIG. **10** is a cross-sectional view of a LDMOS transistor in accordance with some embodiments. The LDMOS transistor shown in FIG. **10** is a modified embodiment of the LDMOS transistor in FIG. **7**. For details, in the LDMOS transistor shown in FIG. **7**, the first active gate stack **120** overlaps with the first region **106** and does not overlap with the second region **108**, and the second active gate stack **126** overlaps with the third region **110** and does not overlap with the second region **108**. However, in the LDMOS transistor shown in FIG. **10**, the first active gate stack **120** partially overlaps with the first region **106** and partially overlaps with the second region **108**, and the second active gate stack **126** partially overlaps with the third region **110** and partially overlaps with the second region **108**.

(45) In accordance with some embodiments of the present disclosure, a MOS transistor includes a substrate, a first region, a second region, a source region, a drain region, an active gate stack, and a dummy gate stack. The substrate has a first conductivity. The first region having the first conductivity is formed in the substrate. The second region having a second conductivity is formed in the substrate, and the second region is adjacent to the first region. The source region with the second conductivity is formed in the first region. The drain region with the second conductivity is formed in the second region. The active gate stack is disposed on the first region. The dummy gate stack is disposed on the second region, and the dummy gate stack is electrically coupled to a variable voltage.

(46) In accordance with some embodiments of the present disclosure, a MOS transistor includes a substrate, a first region, a second region, a third region, a first source region, a drain region, a second source region, a first active gate stack, a first dummy gate stack, a second active gate stack, and a second dummy gate stack. The substrate has a first conductivity. The first region having the first conductivity is formed in the substrate. The second region having a second conductivity is formed in the substrate, and the second region is adjacent to the first region. The third region having the first conductivity is formed in the substrate, and the third region is adjacent to the second region. The first source region with the second conductivity is formed in the first region. The drain region with the second conductivity is formed in the second region. The second source region with the second conductivity is formed in the third region. The first active gate stack is disposed on the first region. The first dummy gate stack is disposed on the second region between the first source region and the drain region, and the first dummy gate stack is electrically coupled to a first variable voltage. The second dummy gate stack is disposed on the second region between the second source region and the drain region, and the second dummy gate stack is electrically coupled to a second variable voltage.

(47) In accordance with some embodiments of the present disclosure, a MOS transistor includes a substrate, a source region, a drain region, a channel region, an extended drift region, an active gate stack, and a dummy gate stack. The substrate has a first conductivity. The source region having a second conductivity is formed in the substrate. The drain region having the second conductivity is formed in the substrate, and the source region and the drain region are separated from each other. The channel region having the first conductivity is formed in the substrate between the source region and the drain region. The extended drift region having the second conductivity is formed in the substrate around the drain region. The active gate stack is disposed on the channel region. The dummy gate stack is disposed on the extended drift region, and the dummy gate stack is electrically coupled to a variable voltage.

(48) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming an isolator on a substrate; forming a channel region in the substrate by implanting a first conductivity type material in the substrate; forming an extended drift region within the substrate by implanting a second conductivity type material in the substrate; forming a source region in between the channel region and the isolator by implanting the second conductivity type material in the substrate; forming a drain region on the extended drift region by

implanting the second conductivity type material in the substrate, wherein a top surface of the drain region is aligned with a top surface of the isolator, a top surface of the source region, a top surface of the extended drift region and a top surface of the channel region; forming an active gate stack including a gate dielectric layer and a gate electrode layer directly on the top surface of the channel region, wherein a sidewall of the gate dielectric layer and a sidewall of the gate electrode layer are aligned with a junction between the channel region and the extended drift region; and forming a dummy gate stack including a dummy gate dielectric and a dummy gate electrode directly on the top surface of the extended drift region, wherein a sidewall of the dummy gate dielectric and a sidewall of the dummy gate electrode are aligned with a junction between the extended drift region and the drain region, and a bottom surface of the dummy gate dielectric is aligned with a bottom surface of the gate dielectric layer, and aligned with the top surface of the drain region, the top surface of the source region, the top surface of the extended drift region and the top surface of the channel region.

2. The method according to claim 1, wherein the source region and the drain region are implanted with the second conductivity type material at a higher concentration than the second conductivity type material implanted in the extended drift region.

3. The method according to claim 1, wherein the active gate stack and the dummy gate stack are formed with the same photomask process.

4. The method according to claim 1, further comprising: forming an insulating layer over the source region and the drain region of the substrate and covering the active gate stack and the dummy gate stack.

5. The method according to claim 4, wherein the insulating layer is formed to cover and contact the extended drift region, the source region and the drain region, and is formed to be physically separated from the channel region.

6. The method according to claim 1, further comprising: forming a second dummy gate stack directly on the top surface of the extended drift region, wherein a sidewall of the second dummy gate stack is aligned with a second junction between the extended drift region and the drain region.

7. The method according to claim 6, wherein the drain region is formed in between the dummy gate stack and the second dummy gate stack.

8. The method according to claim 1, further comprising: forming contact structures connected to the source region, the drain region, the active gate stack and the dummy gate stack.

9. A method, comprising: forming a first channel region and a second channel region in a substrate; forming an extended drift region in between the first channel region and the second channel region; forming a first source region on the first channel region, and a second source region on the second channel region; forming a drain region on the extended drift region; forming a first active gate stack on the first channel region, wherein a first sidewall of the first active gate stack is aligned with a junction between the first source region and the first channel region, and a second sidewall of the first active gate stack is aligned with a junction between the extended drift region and the first channel region; forming a second active gate stack on the second channel region, wherein a first sidewall of the second active gate stack is aligned with a junction between the second source region and the second channel region, and a second sidewall of the second active gate stack is aligned with a junction between the extended drift region and the second channel region; and forming a drain contact electrically connected to the drain region, wherein the drain contact is located in between the second sidewall of the first active gate stack and the second sidewall of the second active gate stack, and wherein a bottom surface of the drain contact is aligned with bottom surfaces of the first active gate stack and the second active gate stack, and aligned with a top surface of the extended drift region.

10. The method according to claim 9 further comprising: forming a first dummy gate stack and a second dummy gate stack on the extended drift region, wherein top surfaces and bottom surfaces of the first dummy gate stack and the second dummy gate stack are aligned with top surfaces and the

bottom surfaces of the first active gate stack and the second active gate stack.

11. The method according to claim 10, wherein a width of the first dummy gate stack and a width of the second dummy gate stack are formed to be smaller than a width of the first active gate stack and a width of the second active gate stack.

12. The method according to claim 10, wherein the first dummy gate stack is formed with a sidewall that is aligned with a first junction between the extended drift region and the drain region, and the second dummy gate stack is formed with a sidewalls that is aligned with a second junction between the extended drift region and the drain region, and the first junction is opposite to the second junction.

13. The method according to claim 10, wherein the first active gate stack is electrically coupled to a first bias voltage, the second active gate stack is electrically coupled to a second bias voltage, the first dummy gate stack is electrically coupled to a first variable voltage and the second dummy gate stack is electrically coupled to a second variable voltage.

14. The method according to claim 9, wherein the drain region is formed to be physically separated from the first channel region and the second channel region.

15. The method according to claim 9, further comprising: forming a first source contact on the first source region, and a second source contact on the second source region, wherein the first active gate stack is located in between the first source contact and the drain contact, and the second active gate stack is located in between the second source contact and the drain contact, and bottom surfaces of the first source contact and the second source contact are aligned with the bottom surface of the drain contact.

16. A method, comprising: forming a transistor, comprising: providing a substrate having a first region with a first conductivity type and a second region with a second conductivity type; forming a source region on the first region, and a drain region on the second region; forming an active gate stack on the first region of the substrate, and electrically coupling the active gate stack to a bias voltage; forming a dummy gate stack on the second region of the substrate, and electrically coupling the dummy gate stack to a variable voltage; and applying the variable voltage to the dummy gate stack to adjust a resistance of the second region under the dummy gate stack, and so that a breakdown voltage of the transistor is adjusted; and forming a second dummy gate stack on the second region of the substrate, and electrically coupling the second dummy gate stack to a second variable voltage.

17. The method according to claim 16, wherein the dummy gate stack is formed with a sidewall that is aligned with a first junction between the second region and the drain region, and the second dummy gate stack is formed with a sidewall that is aligned with a second junction between the second region and the drain region.

18. The method according to claim 17, wherein the second junction is arranged opposite to the first junction.

19. The method according to claim 16, wherein the substrate is further provided with a third region with the first conductivity type, and the method further comprises forming a second active gate stack on the third region of the substrate, and electrically coupling the second active gate stack to a second bias voltage.

20. The method according to claim 16, further comprises: forming contact structures connected to the active gate stack and the dummy gate stack; and forming an insulating layer surrounding the active gate stack, the dummy gate stack and the contact structures.
